

Block Ciphers

Module 3, Part 1: Foundations and Data Encryption Standard (DES)
CSC 3114 Cryptology and Coding Theory

Denish Azamuke, PhD

Office: CoCIS, Block B, Level 3

September 7, 2026

Lecture Outline

- 1 Introduction to block ciphers
- 2 Design principles: confusion and diffusion
- 3 Substitution–permutation and Feistel structures
- 4 Data Encryption Standard (DES)
- 5 DES security lessons and limitations

Module 3

Part 1: Block Ciphers and DES

- Block cipher model
- Confusion and diffusion
- Product ciphers
- Feistel networks
- DES structure
- DES security limitations

Part 2: AES, Integrity, and Streams

- Advanced Encryption Standard
- Secure communication
- Message integrity
- Hash functions
- Message Authentication Codes
- Stream ciphers

Learning Outcomes

By the end of this lecture, you should be able to:

- Explain the block cipher abstraction
- Distinguish block ciphers from stream ciphers
- Describe confusion, diffusion, and avalanche effect
- Explain why Feistel networks are useful in cipher design
- Trace the main stages of DES encryption
- Discuss why DES is no longer sufficient for modern security

From Encryption to Block Ciphers

In Module 2, we discussed encryption as a process for protecting messages.

A **block cipher** encrypts a fixed-size block of plaintext using a secret key:

$$\text{Enc}_K : \{0, 1\}^n \rightarrow \{0, 1\}^n.$$

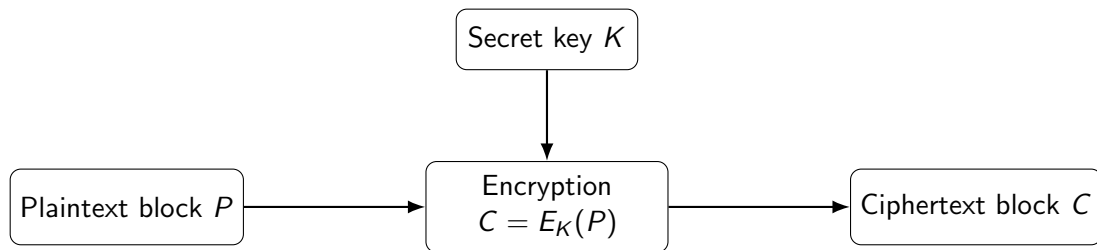
For correct decryption, there must be an inverse transformation:

$$\text{Dec}_K(\text{Enc}_K(P)) = P.$$

Main idea

A block cipher behaves like a keyed permutation over all possible n -bit blocks.

Block Cipher Model



$$P, C \in \{0, 1\}^n, \quad K \in \{0, 1\}^k.$$

Block Ciphers vs Stream Ciphers

Block cipher	Stream cipher
Encrypts fixed-size blocks	Encrypts data one bit or byte at a time
Uses modes of operation for long messages	Generates a keystream
Example: DES, AES	Example: RC4, ChaCha20
Often used for files, disks, and protocols	Often used where low latency is needed

Key distinction

AES and DES are block ciphers. They do not directly encrypt arbitrary-length messages without a mode of operation.

Why Not Just Substitute Symbols?

Classical substitution ciphers replace one symbol with another.

Modern block ciphers must resist attacks even when an adversary can observe many plaintext–ciphertext pairs.

A strong block cipher should make the relationship between plaintext, key, and ciphertext appear complex:

$$(P, K) \mapsto C.$$

Design challenge

Small changes in plaintext or key should cause large, unpredictable changes in ciphertext.

Shannon's Two Design Principles

Claude Shannon identified two important ideas for strong encryption.

Confusion

Make the relationship between the key and ciphertext complex.

Diffusion

Spread the influence of each plaintext bit over many ciphertext bits.

In practice

Modern ciphers repeatedly combine substitutions, permutations, and key mixing.

Avalanche Effect

A cipher has good **avalanche behavior** if changing one input bit changes many output bits. Ideally, flipping one plaintext bit should change about half of the ciphertext bits.

$$P \rightarrow C, \quad P' = P \oplus e_i \rightarrow C'.$$

Expectation

C and C' should look very different, even though P and P' differ by only one bit.

Product Ciphers

A **product cipher** builds a strong cipher by composing many simple transformations.

$$C = T_r \circ T_{r-1} \circ \cdots \circ T_2 \circ T_1(P).$$

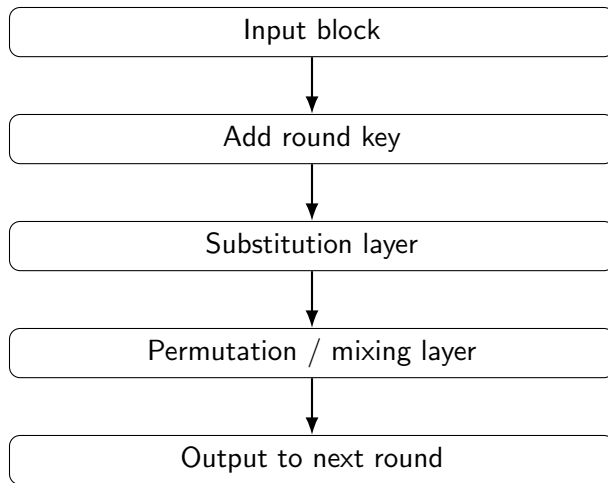
Each round usually performs:

- substitution;
- permutation;
- key mixing.

Reason

One simple transformation may be weak, but many carefully designed rounds can produce strong security.

Substitution–Permutation Network



Repeated rounds gradually hide statistical patterns in the plaintext.

Feistel Network: Main Idea

A Feistel cipher splits the input block into two halves:

$$P = (L_0, R_0).$$

For round i :

$$L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus F(R_{i-1}, K_i).$$

Key advantage

The round function F does not need to be invertible, but the whole Feistel round is still invertible.

Why Feistel Decryption Works

Encryption round:

$$L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus F(R_{i-1}, K_i).$$

Since $L_i = R_{i-1}$, we recover

$$R_{i-1} = L_i.$$

Then

$$L_{i-1} = R_i \oplus F(L_i, K_i).$$

Conclusion

To decrypt, use the same structure but apply the round keys in reverse order.

Data Encryption Standard

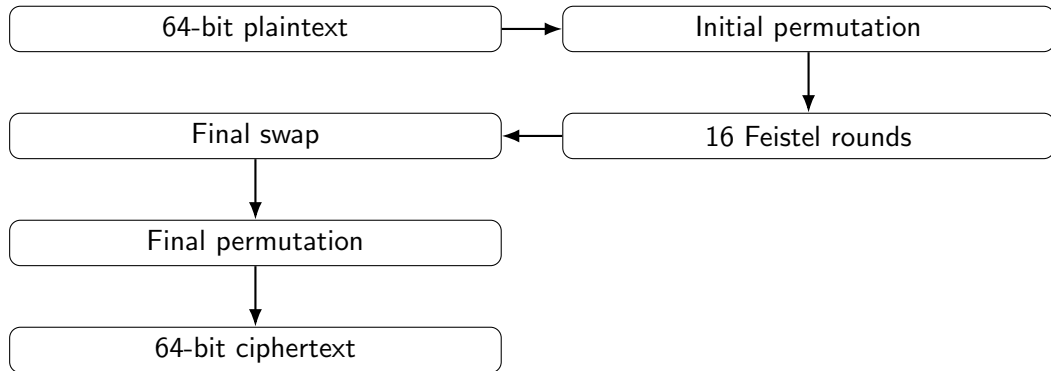
The **Data Encryption Standard (DES)** is a classic Feistel block cipher.

- Block size: 64 bits
- Key input: 64 bits
- Effective key length: 56 bits
- Number of rounds: 16
- Round structure: Feistel network

Historical importance

DES is no longer secure for modern use, but it remains important for learning block cipher design.

High-Level DES Encryption



DES Round Function

For round i , DES uses

$$L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus f(R_{i-1}, K_i).$$

The DES round function f takes:

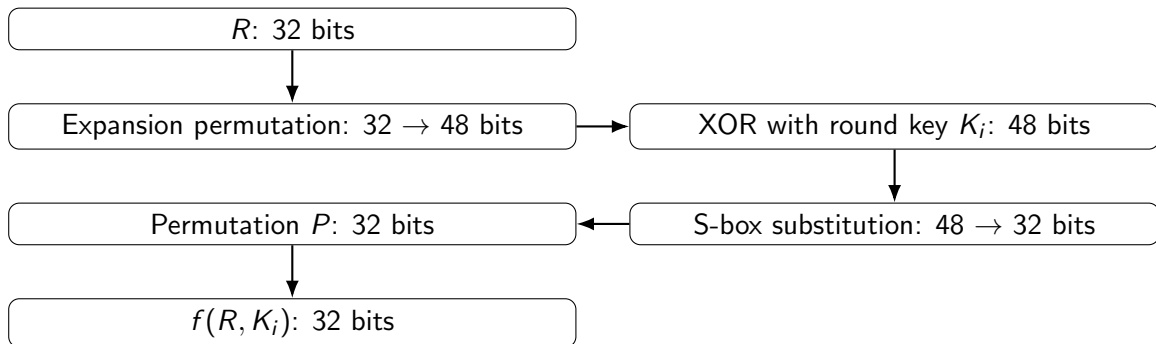
$$R_{i-1} \in \{0, 1\}^{32}, \quad K_i \in \{0, 1\}^{48},$$

and produces a 32-bit output.

Purpose of f

The function f introduces nonlinearity and diffusion inside each round.

Inside the DES Function f



DES Expansion Step

The right half R has 32 bits.

DES expands it to 48 bits before key mixing:

$$E : \{0, 1\}^{32} \rightarrow \{0, 1\}^{48}.$$

- Some bits are repeated.
- Repetition helps spread local changes.
- The expanded block can be XORed with a 48-bit round key.

$$E(R) \oplus K_i.$$

DES S-Boxes

DES uses eight S-boxes:

$$S_1, S_2, \dots, S_8.$$

Each S-box maps 6 input bits to 4 output bits:

$$S_j : \{0, 1\}^6 \rightarrow \{0, 1\}^4.$$

Therefore,

$$8 \times 6 = 48 \quad \text{input bits}, \quad 8 \times 4 = 32 \quad \text{output bits}.$$

Role of S-boxes

S-boxes are the main source of nonlinearity in DES.

DES Key Schedule

DES starts with a 64-bit key input.

After removing parity bits, the effective key length is 56 bits.

The key schedule generates sixteen round keys:

$$K_1, K_2, \dots, K_{16}, \quad K_i \in \{0, 1\}^{48}.$$

Encryption and decryption

Encryption uses K_1 to K_{16} . Decryption uses K_{16} to K_1 .

DES Decryption

DES decryption uses the same Feistel structure.

The only major change is the order of the round keys:

$$K_{16}, K_{15}, \dots, K_1.$$

Why this matters

A single hardware or software implementation can perform both encryption and decryption by reversing the key schedule.

Security of DES

DES was influential, but its 56-bit key is too short for modern security.

A brute-force attacker tries all possible keys:

$$2^{56} \approx 7.2 \times 10^{16} \text{ keys.}$$

This number is large, but not large enough against modern computing resources.

Lesson

Cipher design must consider both mathematical attacks and practical advances in computing power.

Triple DES as a Transitional Solution

Triple DES applies DES three times with multiple keys:

$$C = \text{Enc}_{K_3}(\text{Dec}_{K_2}(\text{Enc}_{K_1}(P))).$$

- It was designed to extend the useful life of DES.
- It is slower than single DES.
- It has largely been replaced by AES in modern systems.

Transition

In Part 2 of the lecture, we shall introduce AES, the modern replacement for DES.

Lecture Summary

- A block cipher is a keyed permutation on fixed-size blocks.
- Strong ciphers use confusion, diffusion, and repeated rounds.
- Feistel networks make invertibility easier to achieve.
- DES is a 64-bit block cipher with a 56-bit effective key.
- DES is historically important but insecure for modern applications.

Questions?